Manasa Ushannagari

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Summary

Aspiring Data Scientist and AI/ ML enthusiast with strong foundations in Python, Machine Learning. Experienced in data preprocessing, model building, and evaluation using Scikit-learn through academic projects. Skilled in SQL, data visualization, and extracting insights from structured datasets. Passionate about applying AI-driven solutions to solve real-world problems

Education

Bachelor of Technology-Civil Engineering, *B.V.Raju Institute of Technology,Medak*
- 75.5%

Diploma - Civil Engineering, *Governament Polytechnic For Womens,Medak.*
- 75%

Skills

Python & EDA
NumPy, Pandas, Data Cleaning, EDA, Matplotlib, Seaborn.

MYSQL
Joins, Aggregations, Queries, TCL, View, Stored Procedure

Machine Learning
Supervised Learning, Unsupervised Learning, Evaluation, Regularization Techniques.

GenAI
Large Language Models, Prompt Engineering, LangChain, RAG.

Advanced Statistics
Descriptive Statistics, Probability Distribution, Data Gathering Techniques, Inferential Statistics.

Power BI
Data Modeling, DAX, Dashboards, Power Query

Deep Learning
ANN, RNN, LSTM, GRU, Text Processing, Embeddings, Transformer.

Projects

Road Vision AI: CNN-Based Road Condition Detection, *Convolution Neural Network*
- Developed a CNN-based road-condition detection system to classify road images into **3 categories**: **Clean Road, Pothole, and Damaged/Cracked Road**.
- Applied transfer learning using the **VGG16** pre-trained model with **ImageNet** weights for road-image feature extraction.
- Prepared labeled road-image data and split it into **training** and **testing** sets for model development and evaluation.
- Trained a **deep-learning classifier** to **automate detection** of potholes and damaged road surfaces, supporting faster road-maintenance decisions.
- Integrated **OpenCV for frame-by-frame** video analysis to detect road conditions from recorded video footage.

Personalized Health Risk Scoring & Insurance Prediction, *Artificial Neural Network*
- Built an ANN-based health risk and insurance premium prediction system using **13 customer health** and lifestyle features with **2 prediction targets**.
- Applied data preprocessing with median/mode imputation, categorical encoding, standardization, power transformation, and an **80:20 train-test split**.

- Developed a neural network with **1 hidden** layer and **54 neurons**, using the tanh activation function and **RMSprop optimizer.**
- Used **Optuna** for hyperparameter tuning, selecting a 0.00165 learning rate, 24.57% dropout, and **L1/L2 regularization** to reduce overfitting.
- Evaluated model performance using training and validation loss and **MAE**; improved generalization with **Dropout**, **Batch Normalization**, and **regularization.**

Swiggy Delivery Time Prediction, *Machine Learning*
- Built a Swiggy delivery-time prediction system using **27 dataset variables**, removing **6 non-predictive** columns and defining delivery time as the regression target.
- Applied median imputation for numerical fields, mode imputation for categorical fields, and One-Hot and Ordinal Encoding for data preparation.
- Split data into **80% training and 20% testing sets** standardized numerical features and applied power transformation to reduce skewness.
- Trained and compared 7 regression models: **Linear Regression, KNN, SVR, Decision Tree, Random Forest, Gradient Boosting, and XGBoost.**
- Selected **XGBoost** Regressor using RMSE, MSE, and R^2 metrics, then deployed the model through a **Streamlit** application for real-time delivery-time predictions

Customer Churn Analysis, *PowerBI*
- Analyzed **7,000+ telecom customer** records to identify churn patterns, revealing an overall churn rate of ~27%
- Built **4+ interactive dashboards** to track KPIs such as churn rate, revenue, customer distribution, and average tenure
- Developed **5+ DAX measures** (Churn Rate, Total Revenue, Average Charges, Average Tenure) to improve reporting accuracy and insights
- Identified **high-risk segments** including month-to-month contract users, fiber optic customers, and electronic check users driving higher churn
- Provided data-driven recommendations (long-term contracts, bundled services, automated payments) to improve **customer retention** and **reduce revenue loss**

Employee Management System, *MYSQL*
- Designed and implemented an Employee Management System database using **MySQL**, creating **6 relational tables** Employee, Job Department, Salary/ Bonus, Qualification, Leaves, and Payroll.
- Established **primary keys** and **foreign key** relationships to connect **5+** employee management modules, ensuring data integrity and efficient database structure.
- Developed **15+** SQL queries using **JOIN, GROUP BY**, and **aggregate functions** to analyze employee distribution, department-wise salaries, and top-paid employees.
- Performed **payroll** and **leave analysis**, generating insights on salary expenditure and employee leave patterns to support **HR decision-making.**

Exploring Housing Trends with Magic Bricks Data, *Exploratory Data Analysis*
- Performed Exploratory (EDA) on a housing dataset using **Python, Pandas, Matplotlib**, and **Seaborn**.
- Analyzed rental prices where most properties were priced below **₹20,000**, with property sizes mainly ranging between **500–2000 sq. ft.**
- Identified that **3-bedroom** properties were the most common, followed by 2-bedroom homes, while **55.2%** of houses were **semi-furnished**.
- Found a strong positive relationship between square footage, bedroom count, and property prices, showing that larger houses generally **have higher prices**.
- Generated meaningful business insights through **data cleaning, statistical analysis**, and **data visualization techniques.**

Certificates

Python Programming
Innomatics Research Labs

Exploratory Data Analysis
Innomatics Research Labs

MySQL
Innomatics Research Labs

PowerBI
Innomatics Research Labs

Machine Learning
Innomatics Research Labs